

Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) combining strong data-analytics skills with practical experience translating ambiguous technical problems into structured, documented solutions at Deutsche Bundesbank (AnaCredit), including stakeholder communication with reporting banks. Hands-on experience with AI-assisted development tools (Claude Code) informs an interest in digital transformation topics, paired with a structured, detail-oriented working style suited to management consulting.

Core Competencies

- **Data Analytics:** Python (Pandas, NumPy, Scikit-Learn), SQL, statistical analysis, data visualization (Matplotlib, Seaborn, Tableau)
- **Structured Problem-Solving:** Translating ambiguous data-quality problems into automated, documented solutions (Deutsche Bundesbank AnaCredit)
- **Stakeholder Communication:** Automated reporting and direct communication with reporting banks on data quality issues
- **Digital Transformation & AI Tools:** Hands-on use of an AI coding assistant (Claude Code) to accelerate a real end-to-end project
- **Programming:** Python, R, SQL, Java, C++
- **MS Office & Tools:** Excel, Word, PowerPoint, Azure Cognitive Services, AWS, Git

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern *04/2026–07/2026*
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to ensure identical business logic after data migration
 - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations), with automated Jira ticket creation and direct email communication to reporting banks
 - Significantly reduced response time to data quality issues

Projects

- **Personal Project (GitHub), built with Claude Code**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Designed and built an end-to-end data pipeline for simulated bank transactions (150 accounts), using Claude Code as an AI coding assistant to accelerate implementation
 - Produced quality visualizations and a structured exception report for stakeholder review
- **Tecnológico de Monterrey**
Predictive Modeling – Healthcare Optimization 10/2024–12/2024
 - Analyzed 25+ years of national health data to identify patient risk profiles and resource bottlenecks; built predictive models and communicated findings through structured visualizations
- **Tecnológico de Monterrey**
Air Quality Analysis Mexico City – SVM Classification 04/2023–06/2023
 - Built SVM models to classify critical ozone levels from citywide data; visualized and communicated findings on regional exposure patterns
- **Tecnológico de Monterrey**
Lattice-Based Cryptography Research 10/2024–02/2025
 - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python; contributed to a scientific publication

Education

- **Tecnológico de Monterrey** **Mexico**
BSc in Data Science 2022–2027
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, numerical optimization.
- **Universität Göttingen** **Germany**
Exchange Semester 2025–2026

Languages

- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.